

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$x = -4$ or $y = -5$	B1	1.1b
	$P(-4, -5)$	B1	2.2a
		(2)	
(b)	$3x + 40 = -2(x + 4) - 5 \Rightarrow x = \dots$	M1	1.1b
	$x = -10.6$	A1	2.1
		(2)	
(c)	$a > 2$	B1	2.2a
	$y = ax \Rightarrow -5 = -4a \Rightarrow a = \frac{5}{4}$	M1	3.1a
	$\{a : a \leq 1.25\} \cup \{a : a > 2\}$	A1	2.5
		(3)	
(7 marks)			

Notes:

(a)

B1: One correct coordinate. Either $x = -4$ or $y = -5$ or $(-4, \dots)$ or $(\dots, -5)$ seen.

B1: Deduces that $P(-4, -5)$. Accept written separately e.g. $x = -4$, $y = -5$

(b)

M1: Attempts to solve $3x + 40 = -2(x + 4) - 5 \Rightarrow x = \dots$. Must reach a value for x .

You may see the attempt crossed out but you can still take this as an attempt to solve the required equation.

A1: $x = -10.6$ or e.g. $-\frac{53}{5}$ only. If other values are given, e.g. $x = -37$ they must be rejected or the $-\frac{53}{5}$ clearly chosen

as their answer. Ignore any attempts to find y .

Alternative by squaring:

$$3x + 40 = 2|x + 4| - 5 \Rightarrow 3x + 45 = 2|x + 4| \Rightarrow 9x^2 + 270x + 2025 = 4(x^2 + 8x + 16)$$

$$\Rightarrow 5x^2 + 238x + 1961 = 0 \Rightarrow x = -37, -\frac{53}{5}$$

M1 for isolating the $|x + 4|$, squaring both sides and solving the resulting quadratic

A1 for selecting the $-\frac{53}{5}$

Correct answer with no working scores both marks.

(c)

B1: Deduces that $a > 2$

M1: Attempts to find a value for a using their $P(-4, -5)$

Alternatively attempts to solve $ax = 2(x + 4) - 5$ and $ax = 2(x + 4) - 5$ to obtain a value for a .

A1: Correct range in acceptable set notation.

$$\{a : a \leq 1.25\} \cup \{a : a > 2\}$$

$$\{a : a \leq 1.25\}, \{a : a > 2\}$$

Examples: $\{a : a \leq 1.25 \text{ or } a > 2\}$

$$\{a : a \leq 1.25, a > 2\}$$

$$(-\infty, 1.25] \cup (2, \infty)$$

$$(-\infty, 1.25], (2, \infty)$$

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$g(0) = 5$	M1	This mark is given for a method to find $g(0)$
	$gg(0) = g(5) = 13$	A1	This mark is given for a correct value for $gg(0)$
(b)	$(x-2)^2 + 1 > 28$ $(x-2)^2 > 27$ $x-2 > 3\sqrt{3}$	M1	This mark is given for a method to solve $g(x) > 28$ when $x \leq 2$
	$x < 2 - 3\sqrt{3}$	A1	
	$4x - 7 > 28$ $4x > 35$ $x > \frac{35}{4}$	M1	This mark is given for a solving $g(x) > 28$ when $x > 2$
	$x < 2 - 3\sqrt{3}$ and $x > \frac{35}{4}$	A1	This mark is given for a correct range of values of x for which $g(x) > 28$ stated
(c)	h^{-1} exists since h is a one-to-one function; g^{-1} does not exist since g is a many-to-one function	B1	This mark is given for a valid explanation
(d)	$h^{-1}(x) = 2 - \sqrt[3]{x-1}$	B1	This mark is given for finding an expression for $h^{-1}(x)$
	$2 \pm \sqrt[3]{x-1} = -\frac{1}{2}$	M1	This mark is given for a method to rearrange to find a value for x
	$x = 7.25$	A1	This mark is given for a correct value of x
(Total 10 marks)			

Q3.

Question	Scheme	Marks	AOs
(a)	$gf(x) = 3 \ln e^x$	M1	1.1b
	$= 3x, (x \in \mathbb{R})$	A1	1.1b
		(2)	
(b)	$gf(x) = fg(x) \Rightarrow 3x = x^3$	M1	1.1b
	$\Rightarrow x^3 - 3x = 0 \Rightarrow x =$	M1	1.1b
	$\Rightarrow x = (+)\sqrt{3}$ only as $\ln x$ is not defined at $x = 0$ and $-\sqrt{3}$	M1	2.2a
		(3)	
(5 marks)			
Notes:			
(a)			
M1: For applying the functions in the correct order			
A1: The simplest form is required so it must be $3x$ and not left in the form $3 \ln e^x$			
An answer of $3x$ with no working would score both marks			
(b)			
M1: Allow the candidates to score this mark if they have $e^{3 \ln x} = \text{their } 3x$			
M1: For solving their cubic in x and obtaining at least one solution.			
A1: For either stating that $x = \sqrt{3}$ only as $\ln x$ (or $3 \ln x$) is not defined at $x = 0$ and $-\sqrt{3}$ or stating that $3x = x^3$ would have three answers, one positive one negative and one zero but $\ln x$ (or $3 \ln x$) is not defined for $x \leq 0$ so therefore there is only one (real) answer.			
Note: Student who mix up fg and gf can score full marks in part (b) as they have already been penalised in part (a)			

Q4.

Question	Scheme	Marks	AOs
	$g(x) = \frac{2x+5}{x-3}, x \geq 5$		
(a) Way 1	$g(5) = \frac{2(5)+5}{5-3} = 7.5 \Rightarrow gg(5) = \frac{2("7.5")+5}{"7.5"-3}$	M1	1.1b
	$gg(5) = \frac{40}{9} \left(\text{or } 4\frac{4}{9} \text{ or } 4.\dot{4} \right)$	A1	1.1b
		(2)	
(a) Way 2	$gg(x) = \frac{2\left(\frac{2x+5}{x-3}\right)+5}{\left(\frac{2x+5}{x-3}\right)-3} \Rightarrow gg(5) = \frac{2\left(\frac{2(5)+5}{(5)-3}\right)+5}{\left(\frac{2(5)+5}{(5)-3}\right)-3}$	M1	1.1b
	$gg(5) = \frac{40}{9} \left(\text{or } 4\frac{4}{9} \text{ or } 4.\dot{4} \right)$	A1	1.1b
		(2)	
(b)	{Range:} $2 < y \leq \frac{15}{2}$	B1	1.1b
		(1)	

(c) Way 1	$y = \frac{2x+5}{x-3} \Rightarrow yx - 3y = 2x + 5 \Rightarrow yx - 2x = 3y + 5$	M1	1.1b
	$x(y-2) = 3y+5 \Rightarrow x = \frac{3y+5}{y-2} \left\{ \text{or } y = \frac{3x+5}{x-2} \right\}$	M1	2.1
	$g^{-1}(x) = \frac{3x+5}{x-2}, \quad 2 < x \leq \frac{15}{2}$	A1ft	2.5
		(3)	
(c) Way 2	$y = \frac{2x-6+11}{x-3} \Rightarrow y = 2 + \frac{11}{x-3} \Rightarrow y-2 = \frac{11}{x-3}$	M1	1.1b
	$x-3 = \frac{11}{y-2} \Rightarrow x = \frac{11}{y-2} + 3 \left\{ \text{or } y = \frac{11}{x-2} + 3 \right\}$	M1	2.1
	$g^{-1}(x) = \frac{11}{x-2} + 3, \quad 2 < x \leq \frac{15}{2}$	A1ft	2.5
		(3)	

(6 marks)

Notes for Question	
(a)	
M1:	Full method of attempting $g(5)$ and substituting the result into g
Note:	Way 2: Attempts to substitute $x=5$ into $\frac{2\left(\frac{2x+5}{x-3}\right)+5}{\left(\frac{2x+5}{x-3}\right)-3}$, o.e. Note that $gg(x) = \frac{9x-5}{14-x}$
A1:	Obtains $\frac{40}{9}$ or $4\frac{4}{9}$ or $4.\dot{4}$ or an exact equivalent
Note:	Give A0 for 4.4 or 4.444... without reference to $\frac{40}{9}$ or $4\frac{4}{9}$ or $4.\dot{4}$

Notes for Question Continued	
(b)	
B1:	States $2 < y \leq \frac{15}{2}$ Accept any of $2 < g \leq \frac{15}{2}$, $2 < g(x) \leq \frac{15}{2}$, $\left(2, \frac{15}{2}\right]$
Note:	Accept $g(x) > 2$ and $g(x) \leq \frac{15}{2}$ o.e.
(c) Way 1	
M1:	Correct method of cross multiplication followed by an attempt to collect terms in x or terms in a swapped y
M1:	A complete method (i.e. as above and also factorising and dividing) to find the inverse
A1ft:	Uses correct notation to correctly define the inverse function g^{-1} , where the domain of g^{-1} stated correctly or correctly followed through (using correct notation) on the values shown in their range in part (b). Allow $g^{-1}: x \rightarrow$. Condone $g^{-1} = \dots$ Do not accept $y = \dots$
Note:	Correct notation is required when stating the domain of $g^{-1}(x)$. Allow $2 < x \leq \frac{15}{2}$ or $\left(2, \frac{15}{2}\right]$ Do not allow any of e.g. $2 < g \leq \frac{15}{2}$, $2 < g^{-1}(x) \leq \frac{15}{2}$
Note:	Do not allow A1ft for following through their range in (b) to give a domain for g^{-1} as $x \in \mathbb{R}$
(c) Way 2	
M1:	Writes $y = \frac{2x+5}{x-3}$ in the form $y = 2 + \frac{k}{x-3}$, $k \neq 0$ and rearranges to isolate y and 2 on one side of their equation. Note: Allow the equivalent method with x swapped with y
M1:	A complete method to find the inverse
A1ft:	As in Way 1
Note:	If a candidate scores no marks in part (c), but - states the domain of g^{-1} correctly, or - states a domain of g^{-1} which is correctly followed through on the values shown in their range in part (b) then give special case (SC) M1 M0 A0

Q5.

Question	Scheme	Marks	AOs
	$15 - 2^{x+1} = 3 \times 2^x$	B1	1.1b
	$\Rightarrow 15 - 2 \times 2^x = 3 \times 2^x \Rightarrow 2^x = 3$ or e.g. $\Rightarrow \frac{15}{2^x} - 2 = 3 \Rightarrow 2^x = 3$	M1	1.1b
	$2^x = 3 \Rightarrow x = \dots$	dM1	1.1b
	$x = \log_2 3$	A1cso	1.1b
	(4)		
	Alternative		
	$y = 3 \times 2^x \Rightarrow 2^x = \frac{y}{3} \Rightarrow y = 15 - 2 \times \frac{y}{3}$	B1	1.1b
	$3y + 2y = 45 \Rightarrow y = 9 \Rightarrow 3 \times 2^x = 9 \Rightarrow 2^x = 3$	M1	1.1b
	$2^x = 3 \Rightarrow x = \dots$	dM1	1.1b
	$x = \log_2 3$	A1cso	1.1b
	(4 marks)		

Notes:

B1: Combines the equations to reach $15 - 2^{x+1} = 3 \times 2^x$ or equivalent e.g. $15 - 2^{x+1} - 3 \times 2^x = 0$

M1: Uses $2^{x+1} = 2 \times 2^x$ or e.g. $\frac{2^{x+1}}{2^x} = 2$ to obtain an equation in 2^x and attempts to make 2^x the subject.

See scheme but e.g. $y = 2^x \Rightarrow 3 \times 2^x = 15 - 2^{x+1} \Rightarrow 3y = 15 - 2y \Rightarrow y = \dots$ is also possible

dM1: Uses logs correctly and proceeds to a value for x from an equation of the form $2^x = k$ where $k > 1$

e.g. $2^x = k \Rightarrow x = \log_2 k$

or $2^x = k \Rightarrow \log 2^x = \log k \Rightarrow x \log 2 = \log k \Rightarrow x = \dots$

or $2^x = k \Rightarrow \ln 2^x = \ln k \Rightarrow x \ln 2 = \ln k \Rightarrow x = \dots$

Depends on the first method mark

This may be implied if they go straight to decimals e.g. $2^x = 3$ so $x = 1.584..$ but you may need to check

A1cso: $x = \log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the y -coordinate

Alternative

B1: Correct equation in y

M1: Solves their equation in y and attempts to make 2^x the subject.

dM1: Uses logs correctly and proceeds to a value for x from an equation of the form $2^x = k$ where $k > 1$

e.g. $2^x = k \Rightarrow x = \log_2 k$

or $2^x = k \Rightarrow \log 2^x = \log k \Rightarrow x \log 2 = \log k \Rightarrow x = \dots$

or $2^x = k \Rightarrow \ln 2^x = \ln k \Rightarrow x \ln 2 = \ln k \Rightarrow x = \dots$

Depends on the first method mark

This may be implied if they go straight to decimals e.g. $2^x = 3$ so $x = 1.584..$ but you may need to check

A1cso: $x = \log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the y -coordinate